

# SPRINTATHON'25

## A SPRINT TO REVOLUTION

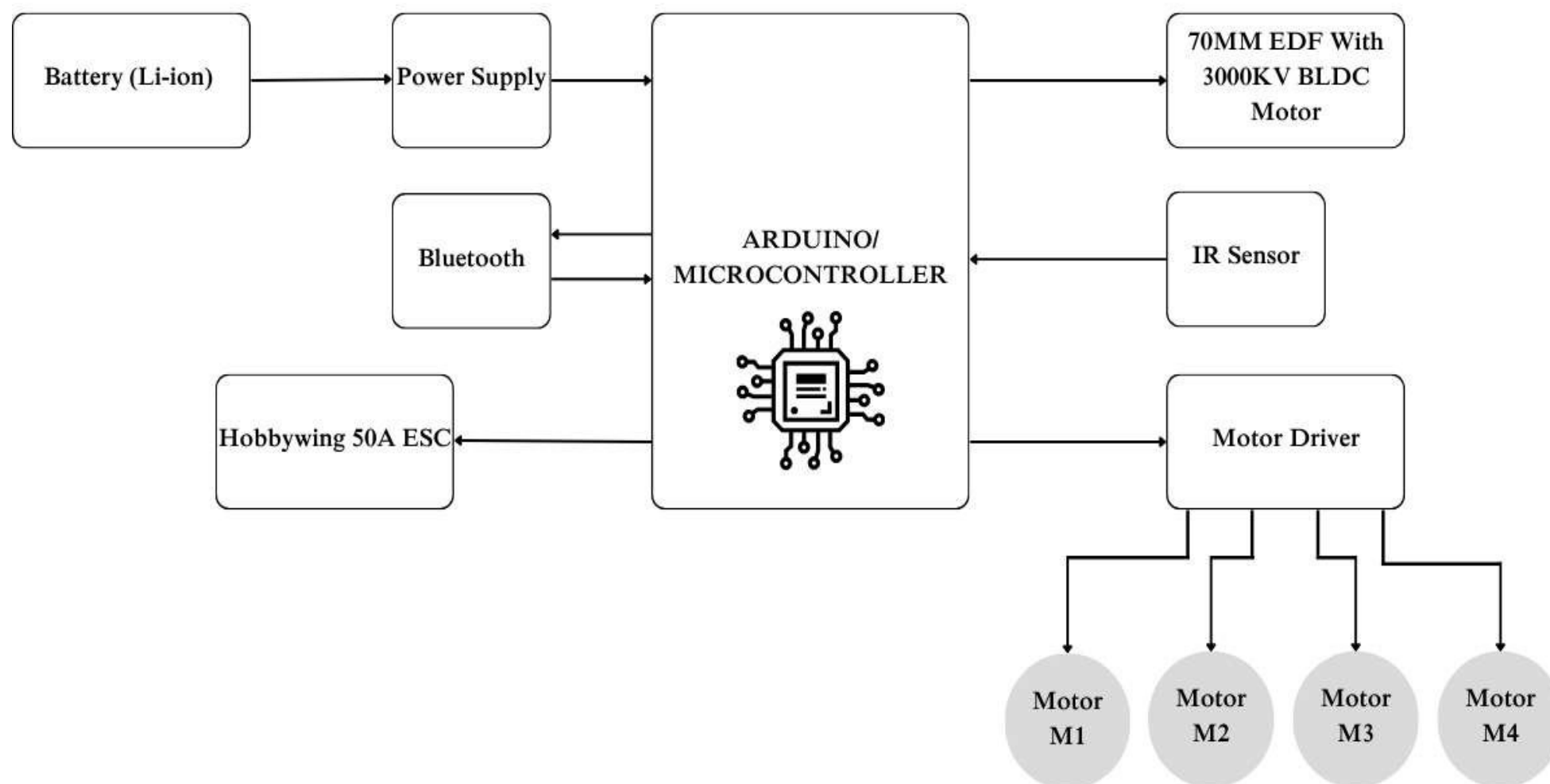
- 1 Problem Statement Number – **PS001**
- 2 Problem Statement – **WALL CLIMBING INSPECTION ROBOT** ( Based on Student Innovation – Encouraging Young Minds to Develop Creative, Technology-Driven Solutions that Address Real-World Challenges, Foster Sustainable Development, and Shape a Smarter, More Connected Future )
- 3 Domain – **Student Innovation**
- 4 Team Details – **404 ERROR SOLVED** ( Team Name )
- 5 Team Members Name – **SURYA.K, VICKKRAMAN.K, SUDESH PILLAI, SANJANA PRAVEEN KUMAR**
- 6 College Name – **SRI VENKATESWARA COLLEGE OF ENGINEERING**

# PROPOSED SOLUTION

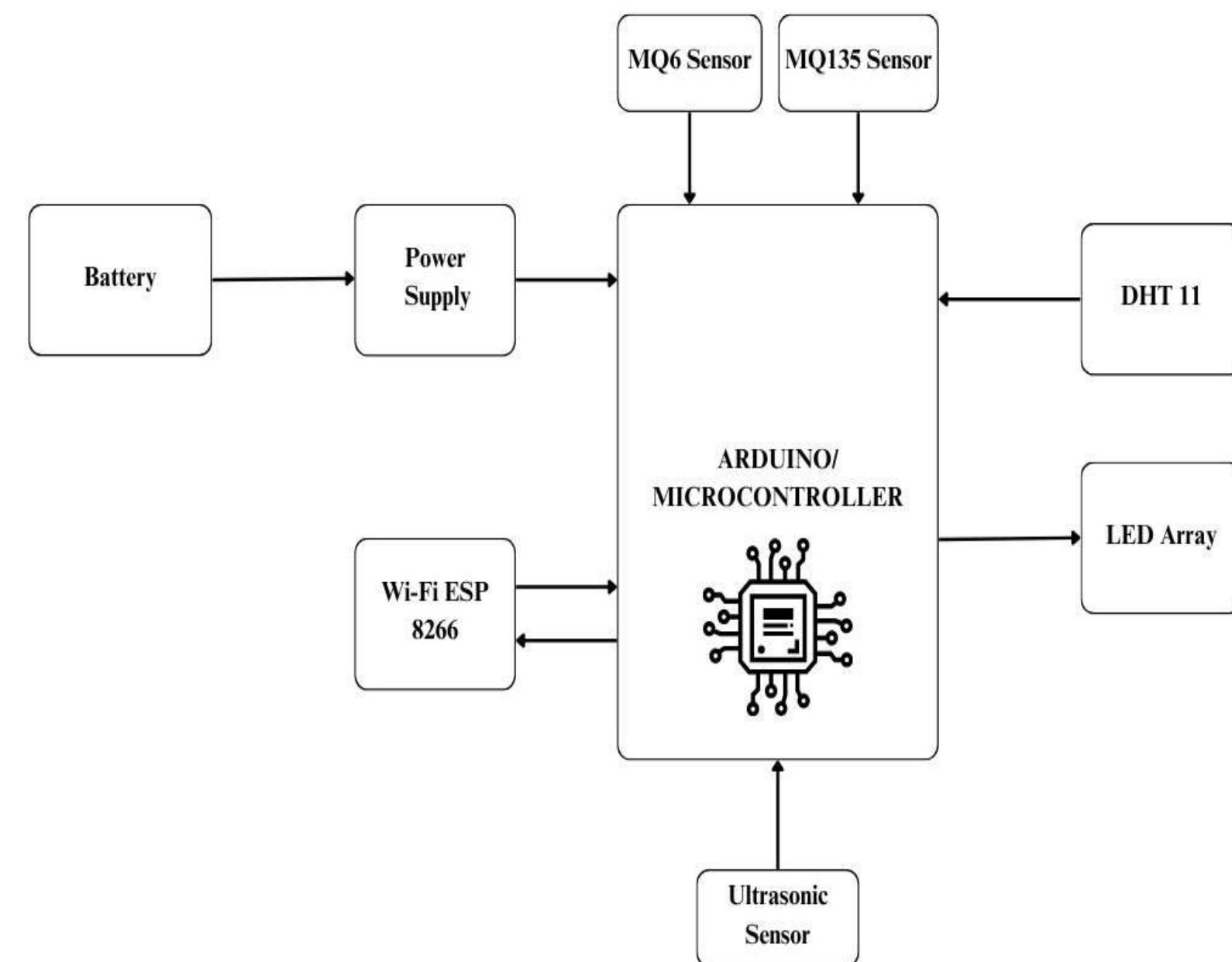
| PROPOSED SOLUTION                                                                                                                                                                                                                                                                                                                                                                                                            | HOW IT ADDRESSES THE PROBLEM                                                                                                                                                                                                                                                                                                                                                                                                                                     | INNOVATION AND UNIQUENESS                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>● A small robot that sticks to walls and crawls safely.</li><li>● Records and monitors the surface in real time.</li><li>● Detects cracks, rust, and leaks during inspection.</li><li>● Shares a live view with the operator.</li><li>● Follows a set path for full surface coverage.</li><li>● Built-in safety features: stops near edges and has a tether to prevent falls</li></ul> | <ul style="list-style-type: none"><li>● Keeps humans away from risky heights (no ropes/scaffolding).</li><li>● Reaches hard-to-access areas quickly (tall buildings, tanks, bridges).</li><li>● Provides consistent checks with digital records for comparison.</li><li>● Saves time and cost compared to manual inspection.</li><li>● Can also be used for crack detection in vehicles such as cars, airplanes, etc and also in nuclear power plants.</li></ul> | <ul style="list-style-type: none"><li>● Uses TinyML to detect the onboard crack and will send alert to the user</li><li>● Unique wall-sticking and crawling mechanism where drones can't work well.</li><li>● Combines visual inspection along with surface sensing in one pass.</li><li>● Creates a defect map/report from raw data.</li><li>● Can adapt to different surfaces (steel, concrete, glass).</li></ul> |

# TECHNICAL APPROACH

- **Programming Languages:** Arduino IDE, Python (for image processing).
- **Frameworks/Tools:** Arduino IDE, TinyML(for crack detection type).
- **Hardware:** Arduino Nano, ESP32-CAM, L298N Motor Driver, BO DC Motors, BLDC/Vacuum Motor, Li-ion Battery, MQ Sensors, DHT11, Ultrasonic Sensor, Wifi ESP8266, LED Array.



Block Diagram of a Wall Climbing Robot



Block Diagram of a Wall Climbing Robot with Sensor Integration

# FEASIBILITY AND VIABILITY

## Analysis of Feasibility of the idea

- **Technological Viability:** Wall-climbing via suction/gripper with lightweight materials; TinyML enables real-time crack detection with low power.
- **Market Demand:** High need for automated, safe inspection of infrastructure; reduces human risk and downtime.
- **Prototype Simplicity:** Open-source hardware/software and ESP32-CAM with deployable TinyML models enable quick integration.
- **Resource Availability:** Low-cost components and free ML libraries make implementation feasible.
- **Scalability & Impact:** Scalable for large-scale monitoring with OTA TinyML updates for long-term accuracy.

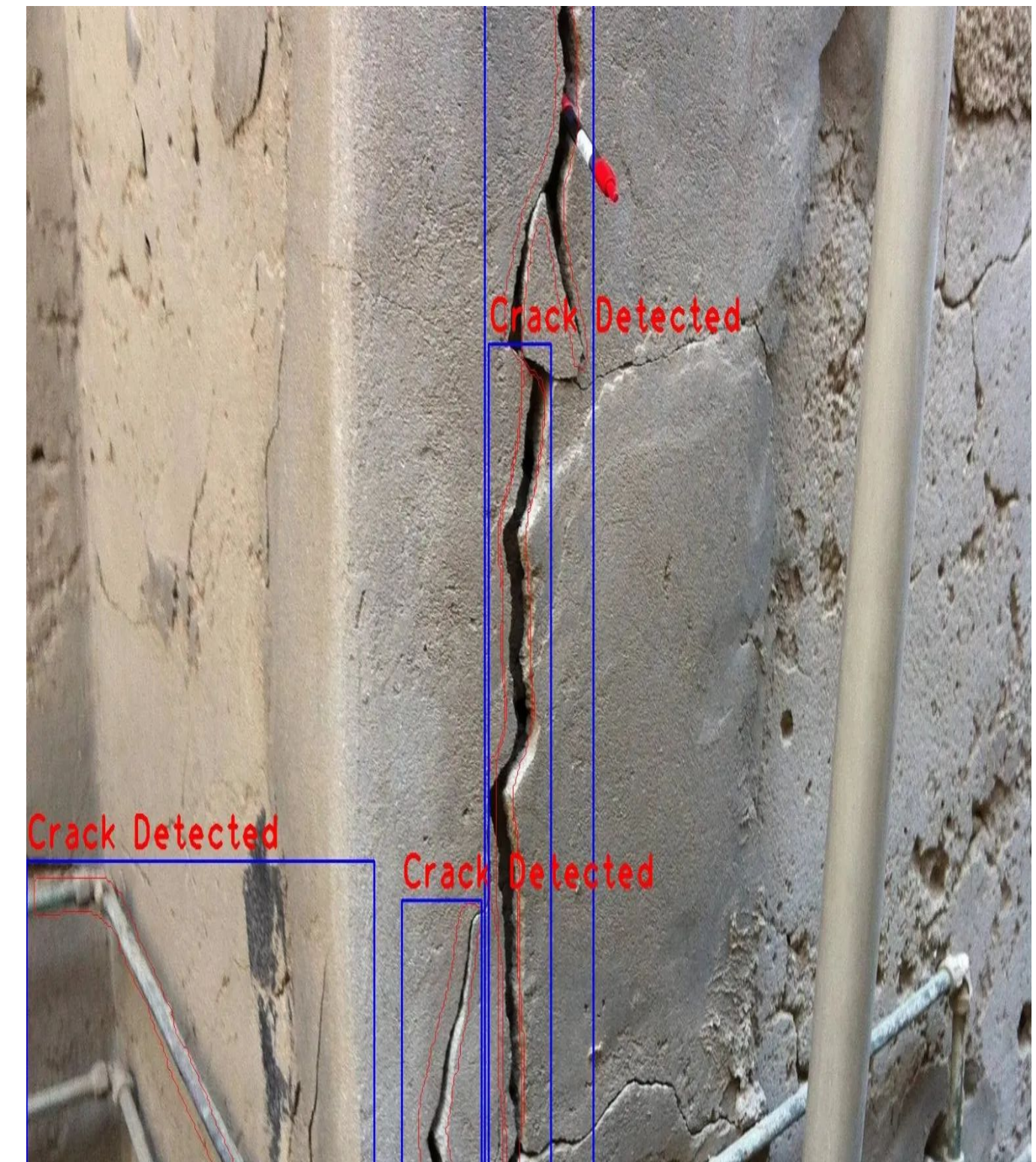
## Potential Risks and Strategies Used

- **Adhesion Reliability:** Risk of slipping on uneven walls → Mitigated by hybrid suction and wheel design and lightweight chassis.
- **Power Limitations:** High energy use from motors and ML tasks → Solved with low-power modes, optimized code, and swappable Li-ion batteries.
- **Model Accuracy:** TinyML may misclassify cracks → Improved by training with diverse datasets and OTA model updates.
- **Environmental Factors:** Dust, poor lighting, or weather interference → Overcome with LED/IR illumination, protective casing, and sensor fusion.
- **Mechanical Durability:** Wear on motors/suction units → Addressed using robust components, modular design, and periodic calibration.



# IMPACT & SOCIAL RELEVANCE

- **Enhanced Safety:** Removes human risk by deploying robots in hazardous inspection zones.
- **Improved Efficiency:** Conducts faster, frequent, and automated inspections, minimizing downtime.
- **Cost-Effective:** Cuts expenses on scaffolding, cranes, and manual labor.
- **High-Quality Insights:** Delivers real-time video, crack classification via TinyML, and environmental sensing for accurate decisions.
- **Versatility:** Adaptable for infrastructure monitoring, solar panel/window cleaning, defense, and disaster response.
- **Sustainability:** Enables early fault detection, lowering repair costs and preventing material wastage.





# MARKET DEMAND ANALYSIS

| TARGET AUDIENCE AND SPECIFIC NEED                                                                                                                                                                                                                                                                                                                                                                                                                                                      | UNIQUE VALUE PREPOSITION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | MARKET POTENTIAL AND SUPPORTING TRENDS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <ul style="list-style-type: none"><li>● Civil infrastructure authorities (bridges, dams, chimneys, high-rises).</li><li>● Power/utility companies requiring periodic inspections of towers and pipelines.</li><li>● Construction and safety regulators aiming to minimize manual inspection risks.</li><li>● <b>Specific need:</b> automated crack detection and structural health monitoring in hard-to-reach areas, reducing dependency on scaffolding and human climbers.</li></ul> | <ul style="list-style-type: none"><li>● Combines <b>wall-climbing mobility</b> (via suction/gripper systems) with <b>on-device TinyML crack detection</b>, unlike drones (limited by battery, turbulence) or manual NDT devices.</li><li>● Uses <b>lightweight, low-cost open-source hardware (ESP32-CAM, Arduino, motors)</b>, ensuring affordability vs. costly industrial robots.</li><li>● Real-time <b>edge inference with TinyML</b> conserves power, removing reliance on cloud processing.</li><li>● OTA (Over-the-Air) update support improves long-term accuracy without hardware replacement.</li></ul> | <ul style="list-style-type: none"><li>● Rising focus on <b>infrastructure safety</b> due to aging bridges/dams and frequent structural failures.</li><li>● Government and smart city initiatives pushing <b>IoT-based monitoring and predictive maintenance</b>.</li><li>● Global inspection robotics market growing due to demand for <b>autonomous, cost-efficient, and risk-free solutions</b>.</li><li>● Trend toward <b>low-power AI at the edge (TinyML, embedded AI)</b> aligns with your robot's design.</li></ul> |

# RESEARCH AND REFERENCES

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- [2] J. Zhu, Y. Zhu, and P. Zhang, "Review of advancements in wall climbing robot techniques," \*Franklin Open\*, vol. 8, Art. no. 100148, Sep. 1, 2024.
- [3] S. Lou, Z. Wei, J. Guo, Y. Ding, J. Liu, and A. Song, "Current Status and Trends of Wall-Climbing Robots Research," \*Machines\*, vol. 13, no. 6, Art. no. 521, Jun. 15, 2025.
- [4] N. Vangala, C. Varunav, J. Aravind, S. M. Yasin, and E. M. Raju, "Design and Analysis of Wall-Climbing Robot," in \*Proc. ICETE 2023: AER 223\*, Lecture Notes in Electrical Engineering, B. Raj et al., Eds., vol. 223, 2023.
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